

Demonstrating Full-hand Electro-Tactile Feedback without Obstructing Palmar Side of Hand

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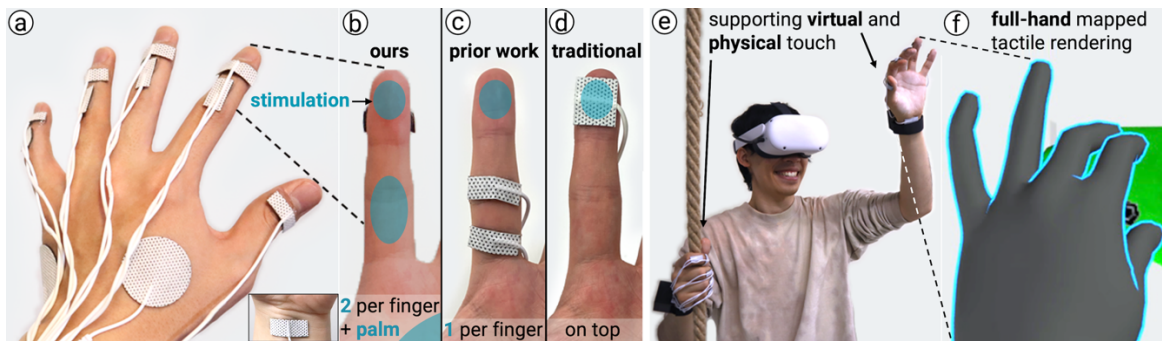


Figure 1: (a, b) We uncover an alternative technique for electro-tactile stimulation that creates touch sensations in 11 distinct locations on the palmar (i.e., front) side of the hand without placing electrodes at the locations where sensations are felt. (c, d) While previous approaches to electro-tactile stimulation showed great promise, they also came with a major limitation as electrodes attached to the palmar side of the hand impair the user’s tactile acuity and dexterity. Instead, we explored placing electrodes in strategic locations at the back of the hand and wrist, which enables the electrical currents to pass through the median/ulnar nerve, thus providing (e, f) tactile feedback to the palmar side of the hand while keeping it free to interact with physical objects.

ABSTRACT

We present a technique to render tactile feedback to the palmar side of the hand while keeping it unobstructed and, thus, preserving manual dexterity during interactions with physical objects. We implement this by applying electro-tactile stimulation only to the back of the hand and to the wrist. In our approach, there are no electrodes on the palmar side, yet that is where tactile sensations are felt. While we place electrodes outside the user’s palm, we do so in strategic locations that conduct the electrical currents to the median/ulnar nerves, causing tactile sensations on the palmar side of the hand.

CCS CONCEPTS

• **Hardware** → Communication hardware, interfaces and storage; Tactile and hand-based interfaces; Haptic devices.

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KEYWORDS

Electro-tactile, Haptics, Mixed Reality, Virtual Reality

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1 INTRODUCTION

A driving vision for HCI has been to bring all our senses into interactive experiences—as Sutherland put it “[the interface] should serve as many senses as possible” [13]. This also implies that haptic devices should not be limited to fingerpads but extend to the whole hand, which is our primary means of interaction. To this end, our community has developed many successful haptic devices that can render contact, texture, or force to *many locations of the user’s palm* [3, 16]. These devices achieve their spatial fidelity by attaching many actuators *directly* to the part of the user’s hand that they are meant to stimulate.

However, **this existing approach to realizing full-hand tactile interfaces comes with a major limitation**. Placing the actuators directly at the user's *palmar side* prevents using these tactile haptic devices beyond purely virtual interactions [9]. In other words, these actuators prevent the user's palms from feeling haptic feedback from the physical object they are grasping, and more dramatically, limit their dexterity to grab or finely manipulate this object—this prevents the usage of these tactile devices in many interactive situations, such as mixed reality.

To circumvent this key issue, researchers have been engineering actuators that minimize the encumbrance in three technical ways. The first is to switch to thin haptic actuators that allow the user to *feel-through* physical objects to some degree [17]. However, the user still feels these thin actuators between their fingerpads and the physical world, resulting in a decreased sensation of textured surfaces [9]. A second alternative is to use *foldable* actuators that keep the user's fingerpad or palms free [15]. These devices fold the actuators away from the palm or the fingerpad when the users are not interacting with virtual objects. However, these are still limited in size (i.e., folding is done by mechanical actuators, which still impair dexterity as these occupy much of the sides and back of fingers/hands) and in the application domain (i.e., do not allow for tactile augmentation of real objects). Finally, a third approach, from which we take inspiration, is to place the haptic actuators away from the fingerpad but drive them in a way that causes the sensation to occur at the fingerpad. Researchers have applied this concept to creating sensations only at the fingerpads, by using vibrotactile actuators on the nail [1], or using electro-tactile stimulation via electrodes on the middle and proximal phalanges of the index finger [18]—unfortunately, these electrodes are *still on the palmar side*. Even a single electrode on any part of the palm would exacerbate manual dexterity. This is because manual activities involve the whole hand [11], and its dexterity builds on the sensory image of the entire palmar side synthesizing each part's sensation [12]. Therefore, we strive to keep the palmar side **entirely unobstructed**.

To address this, we propose applying electro-tactile stimulation *only* to the back of the hand and the wrist—*without electrodes on the palmar side, yet that is where sensations are felt* (Figure 1b). We discovered that this is possible by strategically placing electrodes, building on a neurological concept called *referred sensations*. To find a suitable electrode arrangement, we tackled the following challenges: (1) finding the anatomically suitable location to place electrodes (we found that the back of the hand is superior to the arm, due to the separation of nerves that conduct sensations to each finger); (2) finding a stimulation intensity that creates sensations primarily on the palmar side, rather than the back side (we leverage the hand's dorsal vs. palmar tactile asymmetry); (3) optimizing the stimulation polarities (anodic or cathodic) to increase the number of stimulated locations. The resulting electrode arrangement renders tactile sensations in 11 distinct locations on the palmar side of the hand (please refer to [14] regarding our evaluation).

Finally, enabling haptic feedback to the palm *while keeping the palm free* opens applications where users can feel virtual tactile feedback while *simultaneously* touching and grasping physical objects, which we demonstrated using two applications: (1) integrating

physical props in a VR bouldering experience (Figure 1e), and (2) feeling both virtual & physical models in mixed reality.

2 OVERCOMING THE CHALLENGES TO CREATE TACTILE SENSATIONS AT MULTIPLE POINTS IN THE FRONT OF THE HAND WHILE KEEPING IT ENTIRELY UNOBSTRUCTED

We tackled two, seemingly, incompatible goals: (1) **rendering tactile sensations** on *multiple locations* of the hand's palmar side, while simultaneously (2) **preserving the tactile acuity** of the *same locations* where we create sensations. We tackled this by electrically stimulating the palmar digital nerves (namely, the median and ulnar) via electrodes attached to the dorsal (i.e., back) side of the hand and the wrist. In this section, we illustrate key challenges that led us to find our final electrode arrangement to realize this back-to-front tactile stimulation.

2.1 Decoupling stimulation points from the palmar side of the hand by evoking *referred sensations*

Our approach builds on the fact that electrical currents can cause sensations in mechanoreceptors, even in those distant from the stimulated point—the concept denoted as *referred sensations* [4, 5]. This implies that, in theory, it would be possible to cause sensations on the palmar side of the hand by selectively stimulating nerves anywhere between the hand and the brain (e.g., forearm, elbow, arm). To realize our goal, the first step was to find *suitable stimulation points* that cause sensations at multiple points on the palmar side, including for all individual fingers.

2.2 Challenge #1: stimulation from the forearm or wrist does not realize sensations in individual fingers

Figure 2 (a) shows the two main nerves directly connected to the mechanoreceptors on the palmar side of the hand. As illustrated, both nerves go through the forearm and then the wrist. At the wrist, these nerves connect directly to the palm. This allows us to immediately stimulate the whole palm via electrodes at the wrist, as with prior work leveraging referred sensations to create similar sensations. Theoretically, one would expect to stay at the wrist (or even forearm) and be able to create sensations in the individual fingers. Unfortunately, this is where we found **our first technical challenge—this is not possible as the nerves are too close** to each other (bundled) at the wrist. This prevents (non-implanted) stimulation from evoking sensations in individual fingers without *interference across fingers* [10].

However, as shown in Figure 2 (a), the nerves start *branching out* for the fingers *after the wrist*. Thus, we proposed **stimulating the back of the hand**, where the nerves are separated, becoming easier targets for electrical stimulation; to the best of our knowledge, this is the only way to achieve per-finger referred sensations without interference.

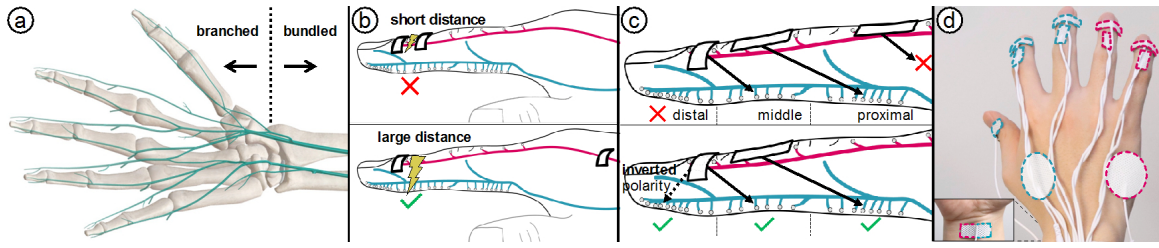


Figure 2: (a) The digital (i.e., finger) nerves branch out for individual fingers after the wrist. (b) An inter-electrode distance must be large to stimulate palmar digital nerves; otherwise, the current flow only stimulates the back side (dorsal digital nerves). (c) Utilizing both polarities (anodic and cathodic) for the electrode attached to the fingertip enables stimulation of the fingerpad. (d) Our final electrode arrangement: we implemented two ground electrodes to avoid median/ulnar interference.

2.3 Challenge #2: electrodes applied naïvely to the back of the hand do not stimulate the front of the hand

Now, one could expect a naïve electrode arrangement for the back of the hand: applying two electrodes (positive and negative) directly atop a target location. Applying this to our technique, one can imagine attaching two electrodes next to each other on the dorsal side of a target finger, hoping that they stimulate the palmar-side mechanoreceptors (Figure 2b). Unfortunately, this is where we find our **second technical challenge—the current flow between these electrodes is too shallow** to stimulate palmar digital nerves and it only stimulates the dorsal side. This is because the closer two electrodes are together, the *shallower* the current penetrates into the tissue [7].

However, as prior work in electrical stimulation has demonstrated, increasing the distance between the electrodes allows the current flow to also stimulate deeper regions [7]. Indeed, adopting this, we discovered that the stimulation with a large inter-electrode distance, as depicted in Figure 2 (b), did generate sensations on the palmar side of the hand.

2.4 Challenge #3: stimulation from the back side creates sensations on *both* sides

Now, the mechanism described so far *should not be sufficient* for realizing our goal. As depicted in Figure 2 (b), the current is stimulating both the dorsal side of the finger (undesired) as well as the palmar side of the finger (desired). Applying this idea naïvely would suggest a **mixed sensation on both sides—our third technical challenge**. To make the sensation on the palmar side more dominant, we leverage the asymmetry in tactile sensitivity between the palmar and dorsal side of the hand. In fact, prior work has shown that **the palmar side has ~60 times more mechanoreceptors (~18000) than the dorsal side (~300)** [8]—this is understandable since the palmar side plays the most important role in dexterous manipulations.

Now, the density of mechanoreceptors is correlated to the sensory threshold to the electro-tactile stimulation [7]. As such, this asymmetry makes the dorsal side less sensitive to stimulation compared to the palmar side. Using this, we can find a stimulation intensity that surpasses the sensory threshold of the palmar side while staying sub-threshold on the dorsal side. With this, we found

it to be feasible to primarily stimulate the palmar side of the hand by adjusting the intensity of electro-tactile stimulation to this range.

2.5 Challenge #4: stimulating the back side of the finger results in offset sensations on the front side

Putting all our solutions so far together, we can posit the electrode arrangement depicted in Figure 2 (c) to create sensations in each finger segment, i.e., distal phalanx (i.e., fingerpad), middle phalanx, and proximal phalanx. Note that these electrodes are paired with a ground electrode attached further away on the back of the hand. We found this arrangement did not perform well for two reasons. Firstly, we found that the proximal-phalanx electrode is not able to cause a tactile sensation on the palmar side—this can be explained by the presence of a tendon layer (central slip) under the electrode location [6]. Instead, we found that the middle-phalanx electrode creates sensations on the palmar side of the proximal phalanx, which revealed a more fundamental issue, our **fourth technical challenge—electrodes on the back of the finger create offset sensations** on the palmar side of the finger, as depicted in Figure 2 (c). Importantly, this revealed that none of these electrodes created sensations at the fingerpad, i.e., the distal-phalanx segment.

To tackle this, we resorted to a known practice demonstrated in prior work: cathodic stimulation advances the location of the sensation in the fingertip direction [18]—yet, so far, all our stimulation followed the conventional anodic current flow. As such, we applied this polarity inversion to the distal-phalanx electrode and confirmed that the electrode is able to stimulate the fingerpad from the back of the finger, resulting in the electrode arrangement depicted in Figure 2 (c). Note that our device uses the distal-phalanx electrode for stimulating the fingerpad and the middle-phalanx targets by switching between the cathodic and anodic stimulation (see the Implementation section for details). Finally, the thumb has only one electrode on the distal phalanx as it has one joint less.

2.6 Challenge #5: stimulating the back side with one ground electrode creates inter-finger interference

Finally, we observed that the resulting electrode configuration so far not only caused sensations in the target finger, but also unwanted tactile feedback in other fingers. This is our **fifth technical**



Figure 3: Our VR bouldering includes grabbing virtual pegs to climb but also manipulating real props (e.g., chalk and a rope).

challenge—fingers do not individually code tactile sensations. For instance, while stimulating a segment of the middle finger, it could also create referred sensations in a segment of the ring or even the pinky finger, which is entirely undesired.

The reason behind this interference is anatomical; Figure 2 (a) depicts how the median and ulnar nerves divide the left and right regions of the hand, i.e., the thumb, index, and middle fingers sense via the median nerve whereas the ring and pinky fingers sense via the ulnar nerve. If an electrical current traverses the two nerves, it can cause sensations to be felt across multiple fingers. To prevent this interference, we implemented two ground electrodes that respectively serve the median and ulnar nerves. By using one of the two grounds at a time depending on which nerve the target finger belongs to, we found that the currents primarily stimulated the correct finger.

Combining our five solutions, described in this section, we established our final electrode arrangement, shown in Figure 2 (d), which can create tactile sensations at up to 15 locations (one on the palm, two on the thumb, three on each of the index, middle, ring, and pinky fingers). Tying this up with experimental results reported in [14], we discourage the use of the middle segments, which we found to exhibit more spatial variance (blurred sensations). Choosing locations where we found lesser spatial variance (focused sensations), our approach totals 11 locations of the user’s palmar side, without placing a single electrode on the palm.

3 APPLICATIONS: KEEPING THE PALMAR SIDE FREE ENABLES NEW INTERACTIVE EXPERIENCES

3.1 Adding Physical Props to VR Bouldering Simulator

Using physical props that stand in for virtual objects is a popular technique to yield higher haptic realism in virtual environments [2]. However, typically, this approach is employed when users are not wearing any haptic wearable devices, such as vibration gloves, because most haptic devices would impair the user’s ability to manipulate and feel the rich tactile feedback from the props. We demonstrate how our technique can alleviate this as it allows users to feel *both* physical props as well virtual feedback. As shown in Figure 3 (a), the user is in a VR bouldering simulator and before climbing the wall, they put chalk onto their hands. Our device does

not encumber the user’s palms while rubbing the real chalk, which acts as a physical prop to enhance immersion.

While climbing, they feel electro-tactile stimulation at the locations where their hands touch the bouldering pegs—in fact, Figure 3 (b) shows how the user’s thumb and pinky are not touching the peg and thus do not experience feedback; this depicts how our approach is selective at the level of individual finger segments.

Moreover, this VR climbing experience is furnished with another prop, a climbing rope attached to a pulley in the ceiling. As depicted in Figure 3 (d), the user can grab the rope and feel its texture, or even apply force on it.

3.2 Interactively Guided Clay Modeling in Mixed Reality

We demonstrate the usage of our system for guiding the user as they model physical shapes in mixed reality (MR) using real clay. This is inspired by prior work on interactive sculpting [19], but takes these ideas further by allowing users to physically model directly with their hands, rather than through handheld tools. Figure 4 (a) depicts the user pushing MR buttons to browse through 3D models to choose which model they want the system to assist them with—as they press buttons, they feel electro-tactile feedback to confirm the actions and add realism to the interactions.

The user now chooses the bear model and attempts to clay a copy of it. To understand the model’s geometry, the user can touch, grasp, move, and rotate the virtual bear with their whole hand. As they do so, our system renders each of the 11 possible palm segments that are in contact with the 3D model, e.g., feeling on the whole hand as they grab the model as shown in Figure 4 (b). Then, as depicted in Figure 4 (c), the user molds clay, putting force on their palm and fingerpads—since our electrodes are attached to the back of the hand and the wrist, they provide minimal encumbrance. Finally, to give the finishing touches to the clay bear’s head, the user aligns the physical and real model and uses our electro-tactile stimulation to feel the places where the VR model differs from the clay model, as depicted in Figure 4 (d).

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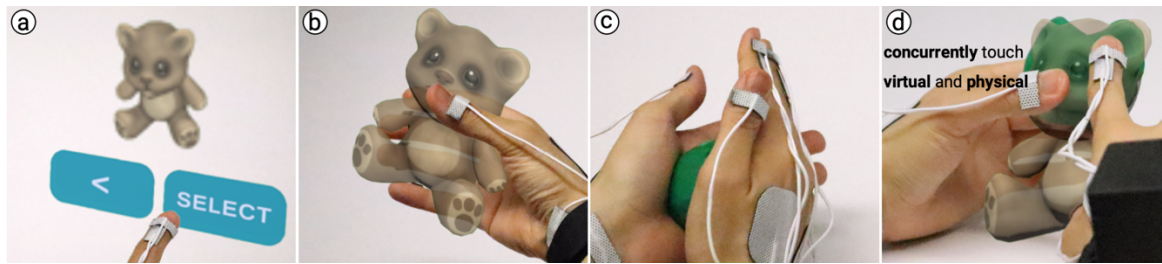


Figure 4: MR clay application alongside virtual 3D models (superimposed the virtual content for the sake of visual clarity).

2047189. Any opinions, findings, and conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of any funding agencies. We are grateful for the feedback from our reviewers.

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